III SEMESTER	L	T	P	C
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20AM3L01 : DATA STRUCTURES AND ALGORITHMS LAB				

COURSE OBJECTIVES:

1. To implement linear and non-linear data structures.
2. To understand the different operations of linear and linear data structures.
3. To implement linear search and binary Search.
4. To get familiarized to sorting algorithms.
5. To exploit appropriate algorithm for the problem.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1: Remembering the concept of data structures through ADT including List, Stack and Queues and its operations.

CO2: Understand the operations on Trees and Graphs.

CO3: Able to apply and implement various sorting algorithms and ensure their correctness.

CO4: Ability to analyze algorithms and develop algorithms through step-by-step approach in solving problems with the help of fundamental data structures.

CO5: Understand the concepts of various Algorithm Design techniques. Design applications and justify use of specific linear data structures for various applications

CO6: Apply algorithms for real time problem solving

LIST OF EXPERIMENTS

1. Implement operations on Array using C.
2. Implement operations on a Singly linked list using C.
3. Implement operations on a Doubly linked list using C.
4. Implement a Stack using an Array and Linked list using C.
5. Convert an infix expression to postfix expression using C.
6. Implement Queue using an Array and Linked list using C.
7. Implement Circular Queue using C.
8. Implement display elements of a Queue according to their priority using C.
9. Search an element using Linear and Binary Search using C.
10. Binary search tree implementation using linked list and possible operations on binary search trees using C.
11. Implement the concepts of In-order, preorder and post order traversals using C.
12. Implement Depth first and Breadth first traversal in graphs using C.
13. Sort the elements using insertion sort using C.
14. Sort the elements using quick sort using C.
15. Sort the elements using merge sort using C.

TEXT BOOKS:

3. Jean-Paul Tremblay, Paul G. Sorenson, ‘An Introduction to Data Structures with Application’, TMH, 2017.
4. Richard F, Gilberg, Forouzan, “Data Structures”, Cengage, 2nd Edition, 2004.

REFERENCE BOOKS:

3. Larry R. Nyhoff, ADTs, Data Structures, and Problem Solving with C++, Prentice Hall Edition, 2004.
4. Thomas H. Cormen, Charles E. Leiserson, “Introduction to Algorithms”, 3rd Edition, 2010.